QP CODE: 25900296

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course(DSC)–MG3DSCCND201- LIFE CYCLE AND WELLNESS NUTRITION-1

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest (I) and Appreciation(Ap)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer all questions
Each question carries **1** mark

1. Option B-Seasonal foods
2. Option B - 300mg
3. Option D-Excessive intake of vitamin C
4. Option A-LBW babies
5. Option D- Limiting junk food and sugary drinks
6. Option B- Breast and ovarian cancer

Part B
Short Answer Type Questions.
Answer any **2** questions.
Each question carries **2** marks.

7.Explain what is meant by nutritional requirement in meal planning.

- A good menu will provide adequate calories, proteins, fats, vitamins and minerals essential for the physical wellbeing of each member of the family. **(1 mark)**
- In a balanced diet the ratio of energy distribution from carbohydrate, protein and fat would be 7:1:2 respectively. **(1 mark)**

8. List two immunological components of colostrum [K]/[CO2]

- Colostrum contains an interferon like substance which possess strong antiviral activity
- It contains B12 binding protein which renders B12 unavailable for the growth of E.coli and other bacteria
- It also contains antibodies against viral diseases such as small pox,polio,measles and influenza **(2 Mark for 2 points)**

9.Expalin why is vitamin C important in preschool nutrition.

- It aids in the absorption of non-haem iron. **(1 mark)**

- It helps in building immunity and protects preschool children from infections. **(1 mark)**

10.Name any four forms of PEM.

- Kwashiorkor
- Marasmus
- Marasmic Kwashiorkor
- Nutritional dwarfing
- Underweight

(Any 4 points and each one carries 0.5 mark)

Part C

Short Essay Type Questions.
Answer any**4** questions.
Each question carries **5** marks.

11. Discuss about how can poor dietary habits in early childhood lead to iron deficiency anaemia in preschoolers.

- **Low intake of iron-rich foods**- Diets lacking in iron sources reduce the amount of iron available for red blood cell production.
- **Early use of cow's milk**- displace iron-rich foods from the diet and also interfere with iron absorption.
- **Diets low in vitamin C** - reduce iron absorption.
- **Inadequate complementary feeding**- Delayed introduction of iron-rich complementary foods after 6 months of can lead to depletion of iron stores accumulated during infancy.
- **Frequent infections and poor overall nutrition** - Malnutrition and repeated infections can impair nutrient absorption and increase iron losses, worsening anaemia in children with already poor diets.

(Each point carries 1 mark)

12.List five suitable low-cost weaning foods and their nutritional benefits[K]/[CO2]

- Ragi Porridge-Rich in calcium and iron, supports bone health, contains complex carbohydrates and fibre for energy
- Mashed Banana-Source of potassium and fibre. Improves digestion and provides energy
- Amutham Porridge- a balanced combination of energy, protein, vitamins, and minerals at a very low cost. It helps combat malnutrition and supports the healthy growth and development of infants and young children.
- Mashed vegetables (carrot, pumpkin, potato, sweet potato)- Supply vitamins (A, C), minerals, and fibre; promote vision, immunity, and bowel regularity.
- Khichdi (rice + moong dal + vegetables)- Balanced combination of carbohydrates, proteins, and vitamins; promotes overall growth and development.

13. Explain the need for micronutrients (iron,vitamin A,iodine)in preventing common deficiencies in school children. [U]/[CO3]

Iron

- The average iron requirement for 10-12 year old girls is 28 mg and for boys 16 mg.
- The RDA of iron for 7-9 year old is 15 mg.
- Iron deficiency anaemia is a risk factor for poor educational performance in school age children **(2 marks)**

Vitamin A

- The RDA of Vitamin A for 7-9 years is 630 µg.
- For 10-12 year old the RDA for boys is 770 µg and for girls 790 µg
- Deficiency of Vitamin A may lead to eye disorders **(2 marks)**

Iodine

- The RDA suggested by ICMR for iodine is 90 µg and 100 µg for 7-9 year old and 10 -12 year old respectively.
- Iodine deficiency can significantly impact the growth and cognitive development of school-aged children**. (1 mark)**

14. Explain the principles of growth and development.

1.All growth and development follow an orderly sequence. A chid can sit only when the muscles of back are ready to support the body.

2. Each child normally passes through a number of stages, each with its own essential characteristics.

3. There are individual differences in the development. Children start walking by the time they are one-year-old but some are early and others are late.

4. Though the human being develops as a unified whole, each part of the body develops at different rates. Basically there are two sequences in the rate of development.

A) Cephalocaudal: Development proceeds from head to toe. The head and brain develops first, then the trunk.

B) Proximodistal: Development proceeds from centre to extremities.

5. Development is essentially the result of the interaction between maturation and learning.

(Each point carries 1 mark)

15. List five food items commonly included in a healthy packed lunch.[K]/[CO3]

- Vegetable pulao,boiled egg,tomato raita and orange fruit
- Fenugreek leaves stuffed chapathi,thick dal and butter milk
- Sambar rice,amaranth porial,buttermilk
- Cheese sandwiches or paneer sandwiches,guava
- Curd Rice,tomato or cucumber slices and coriander chutney.

16. Describe the role of AR formulas in the management of gastroesophageal reflux in infants [K]/[CO2]

- Anti-Regurgitation(AR) Formulas are specially designed infant formulas used to reduce frequent regurgitation or mild reflux in babies**.(1 mark)**
- They are thickened infant formulas that help keep the milk in the stomach,reducing the likelihood of it coming back up. **(1 mark)**

- There are thickening agents that you can add to your regular formula and also specifically formulated formulas that are already thickened. Both tend to stay in the stomach and not escape so easily back up in the oesophagus. **(1 mark)**
- There are many anti-regurgitation/anti-reflux formulas on the market,using different starches as the thickening agents for eg: S26 AR,Novolac AR,NAN AR etc.By adding starch ,it causes the feed to thicken(increases the formulas viscosity)once it reaches the stomach,therefore reducing the possibility of regurgitation **(1 mark)**
- Different thickening agents,such as processed rice,corn or potato starch,guar gum or locust bean gum are used**.(1 mark)**

Part D

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17.Explain the factors affecting growth and development of a child under 6 years.

1. Heredity: It is the process by which features and characteristics are passed from parents to the child before the child is born. Thus features like the colour of the skin and eyes. the height. body build and intelligence are also fixed and no one can change them beyond a limit. **(1.5 marks)**

2. Prenatal environment: This is the environment of the foetus in the womb. If the mother gets poor nutrition, is emotionally upsets, smokes or drinks, takes some medicine or suffers from certain diseases, the growth of the child can be adversely affected. **(1.5 marks)**

3. Nutrition: proper nutrition is essential for the healthy development of the child. A malnourished child's growth is retarded or slow. **(1 mark)**

4.Intelligence: Higher intelligence is associated with faster development while lower intelligence is associated with retardation in various aspects of development. **(1 mark)**

5.Emotional climate at home: If there is a lot of discord or fight at home, the child is not given enough love and attention at home or there is physical or mental abuse of the child, then the child's development is adversely affected. **(1 mark)**

6. Health of the child: If the child frequently falls sick, suffers from some disorder, is disabled or has disturbed endocrine functioning, the development is likely to suffer.
(1 mark)

7. Level of stimulation: The level of stimulation the environment provides to the child-the opportunities for exploration of the environment, opportunities for interaction with other people, all influence the rate of development. **(1 mark)**

8. Socio economic status: It also influences the development by deciding the kind of nutrition, stimulation, facilities, opportunities the child gets. **(1 mark)**

9.Sex: All children follow the same sequence of development. However certain skills are faster in girls than in boys and some other skills are faster in boys than girls.
(1 mark)

18.Describe the factors to be considered in planning a diet.

1.Meeting nutritional requirement: A good menu is one which will not only provide adequate calories, fat and proteins but also minerals, vitamins essential for the physical well-being of each member of family. In a balanced diet the ratio of energy distribution from carbohydrate, protein and fat would be 7: 1: 2 respectively. The diet should contain 'Basic four food groups'.

2. Meal pattern must fulfil family needs: Meal pattern varies with age, occupation and life stage of the family members. A family meal should cater to the needs of the different members. A growing adolescent boy may need variety of foods to satisfy appetite, whereas a young child may require soft and bland diet. Pregnant women require more greens in the diet. A heavy worker requires more calories and B vitamins than other members of the family. The family meal must offer children enough fat and flexibility in caloric density so that their energy needs are met.

3. Meal planning should save time and energy: Planning of meals should be done in such a way, that the recipes should be simple and nutritious. Labour and time saving devices can be used. Using convenience foods save time and energy.

4.Meal planning should consider economic constraints: Meals planned that are not within budget, cannot be put into practice. The cost of meals can be reduced by bulk purchasing and using seasonal fruits and vegetables.

5. Meal plan should give maximum nutrients: Losses of nutrients during processing and cooking should be minimised. Sprouted grams, malted cereals and fermented foods enhance the nutritive value. Good quality protein should be distributed in all meals. Pressure cooker can be used to conserve the nutrients.

6. Meal planning considers individual likes and dislikes: The meal plan should not only meet RDA but also individual preferences, like vegetarian or non-vegetarian preferences. Religion, traditions and customs of the individual should be considered in planning the menu**.**

7. Meals planned should provide variety: If meals are monotonous they are not relished. Variety can be introduced in colour, texture and taste, by using different kinds of foods and cooking methods. Variety also helps in meeting the nutritional requirement.

8. Meals should give satiety: Each meal should have some amount of protein, fat and fibre to give satiety. While planning menu, the interval between meals is also considered.

9. Menus should include locally available foods: The wide variation in dietary patterns throughout the world is due to the availability of different kinds of foods in those regions.

10. Meals should include health foods: Ideal diets should provide besides nutrients those bioactive chemicals which can help to prevent and retard disease processes.

(Each point carries 1 mark)

19. List and explain the common health hazards of bottle feeding in infants [K]/[CO2]

- **Infectious Morbidity:** Compared with breast fed infants,formula fed infants face higher risk of infectious morbidity in the first year of life.These differences in health outcomes can be explained by specific and innate immune factors present in human milk.
- **Ear Infections(Otitis Media):** Approximately,44 % of infants will have atleast 1 episode of otitis media in the first year of life,and the risk among formula-fed infants is doubled compared with infants who are exclusively breast fed for more than 3 months.
- **Lower Respiratory Tract Infection**: Infants who were not breast fed face an increased risk of hospitalization for lower respiratory tract infection in the first year of life,compared with infants who were exclusively breast fed for more than 4 months.The majority of respiratory hospitalizations for infants result from infection with RSV.
- **Gastrointestinal Infections:**Formulated infants face an increased risk of Gastroenteritis and diarrhoea.
- **Necrotizing Enterocolitis :** Among pre-term infants,not being breast fed is associated with an increased risk of Necrotizing Enterocolitis.
- **Obesity and Metabolic Disease:** Children who are formula-fed in infancy are more likely to become obese or develop type 2 diabetes.Children formula-fed in infancy are 1 to 3 times likely to become obese as children who had ever been breast fed.Being

formula fed is associated with an increased risk of type 2 diabetes compared with being breast fed.

- **Neurodevelopment:** Lower IQ scores are reported in formulated children compared with breast fed children
- **SIDS(Sudden Infant Death Syndrome):** Formula feeding is associated with a higher risk of infant mortality compared with breast feeding
- **Infant Mortality :** Formula feeding is associated with an increased risk of infant mortality compared with breast feeding
- **Asthma:**There is a 1.7 fold risk of developing asthma among formula fed children with a positive family history of asthma and a 1.4 fold risk of developing asthma among formula fed children without a family history,compared with those who were breast fed for 3 months or more
- **Atopic Dermatitis :** Infants with a family history of atopy who were exclusively breastfed for less than 3 months have a 1.7 fold risk of atopic dermatitis compared with infants who were exclusively breastfed.
- **Type 1 Dermatitis :** Less than 3 months of breast feeding has been associated with a 1.2 to 1.4 fold increased risk of developing type 1 diabetes compared with more than 3 months of breast feeding.
- **Childhood Cancer:**Several studies have examined associations between formula feeding and childhood leukemia based on the hypothesis that immunoreactive factors in breast milk may prevent viral infections implicated in the leukemia pathogenesis.**(1 mark for each point)**

20. Explain the causes and symptoms of VAD

Causes

1. Maternal vitamin A deficiency: Maternal vitamin A deficiency resulting in low concentrations of vitamin A in breast milk.
2. Inadequate dietary intake of vitamin A: Inadequate dietary intake of vitamin A or its precursor (Beta carotene) during and after weaning.
3. Delayed supplementary feeding: Due to female illiteracy and ignorance supplementation of vitamin A rich foods delayed.
4. Sources of vitamin A rich foods are avoided with the belief that they are deleterious to the health of the children.
5. Poverty
6. Infections and Infestations: The common childhood infections like measles, diarrhoea, respiratory tract infections and parasitic infestations interfere with the absorption of vitamin A.
7. Lack of awareness of the community services: Unaware of PHC services like immunization, vitamin A supplementation and other health services.

(Each point carries 1 mark)

Symptoms

- Dry /thick/scaly skin
- Dry lips
- Dry eyes
- Infertility
- Growth retardation

- Throat and chest infections
- Poor wound healing
- Impaired immunity
- Night blindness
- Conjunctival xerosis
- Bitot's spot
- Corneal xerosis
- Keratomalacia

(Any 6 symptoms and each one carries 0.5 Mark)